

Ahmed Tawfik

Artificial Intelligence · Computer Vision · Deep Learning

✉ ahmed.tawfik96@gmail.com 📍 Kaiserslautern, Germany 📄 [ahmed-tawfik-aboukhadra](#)
🌐 ATAboukhadra 🎓 Google Scholar 🌐 [ataboukhadra.github.io](#)



EDUCATION

Ph.D. Computer Science
RPTU Kaiserslautern, Germany
2021 – Present
3D reconstruction of hand-object interaction.
Supervisor: Prof. Didier Stricker.

M.Sc. Artificial Intelligence
Maastricht University, Netherlands
2019 – 2021
GPA 8.33/10; €58k merit scholarship.

B.Sc. Computer Science & Eng.
German University in Cairo, Egypt
2014 – 2019
GPA 1.06 (German scale, ≈ A).

SKILLS

Python	• • •
PyTorch	• • •
NumPy	• • •
Computer Vision	• • •
3D Reconstruction	• • •
Deep Learning	• • •
Transformers	• • •
Java	• • •
LLMs / RAG	• • •
VLMs	• • •
Git & Linux	• • •
TensorRT / ONNX	• • •
C / C++	• • •
Docker	• • •

Also: OpenCV, HuggingFace, Gaussian Splatting, Pose Estimation, SLURM / HPC.

LANGUAGES

English	• • •
Arabic (native)	• • •
German	• • •

AWARDS

- **1st Place**, ARCTIC Challenge, ICCV 2025.
- UM Holland High-Potential Scholarship (€58k), 2019.
- ECPC & ACPC Finalist, 2017.

PROFILE

Researcher completing a PhD in computer vision, with a broad enthusiasm for AI: from perception and 3D vision to language models, vision-language models, and generative systems. I enjoy taking ideas all the way to practical, working software, from data and model training on GPU clusters to optimised real-time deployment. Author of five publications and a 1st-place result at an international challenge.

EXPERIENCE

Researcher — DFKI Augmented Vision Oct 2021 – Apr 2026
Kaiserslautern, Germany

- PhD research pushing the state of the art in hand-object interaction reconstruction: 3D pose, shape, and geometry from images, depth, and video.
- Published five papers — including at **CVPR Findings** and **WACV** — on Gaussian splatting, transformer/graph networks, and real-time tracking, and won **1st place** at the ARCTIC Challenge (ICCV 2025).
- Built complete pipelines from data preprocessing and model training on GPU clusters through to real-time deployment.
- Co-supervised master’s students and gave guest lectures at RPTU.

Research Intern — CISP Helmholz Center Mar – Aug 2021
MSc thesis: knowledge distillation as a defence against inference attacks on deep CNNs.

ML Intern — HackerOne Sep 2020 – Jan 2021
Built a BERT-based duplicate report detector; ~5% accuracy gain over the heuristic baseline.

Deep Learning Intern — Ulm University Mar – Aug 2018
Bachelor thesis: Arabic handwritten character recognition with CNNs (Python, TensorFlow).

PUBLICATIONS

GHOST: Fast Category-Agnostic Hand-Object Interaction Reconstruction from RGB Videos using Gaussian Splatting. *CVPR Findings 2026*.

Fast-HaMeR: Boosting Hand Mesh Reconstruction using Knowledge Distillation. *VISAPP 2026*.

SurgeonNet: Realtime 3D Pose Estimation of Articulated Surgical Instruments from Stereo Images. *GCPR 2024*.

ShapeGraFormer: GraFormer-Based Network for Hand-Object Reconstruction from a Single Depth Map. *IEEE Access 2024*.

THOR-Net: End-to-End Graformer-Based Realistic Two Hands and Object Reconstruction with Self-Supervision. *WACV 2023*.

PROJECTS

Real-time Hand Mesh Demo: Live 3D hand reconstruction with ONNX/TensorRT acceleration and 3D rendering.

Voice-note assistant: Side project that transcribes voice notes (Whisper) and extracts tasks and notes with an LLM; FastAPI backend.

Educational video generator: Side project that assembles short narrated videos from generated text, images, and voiceovers.